DP Biology – D2.3 (HL extension)

# Water potential

- What affects whether water moves into or out of a cell?
- In what ways do plant cells and animal cells control water movement differently?

HL statements D2.3.8–11. Textbook pp. 690–692.

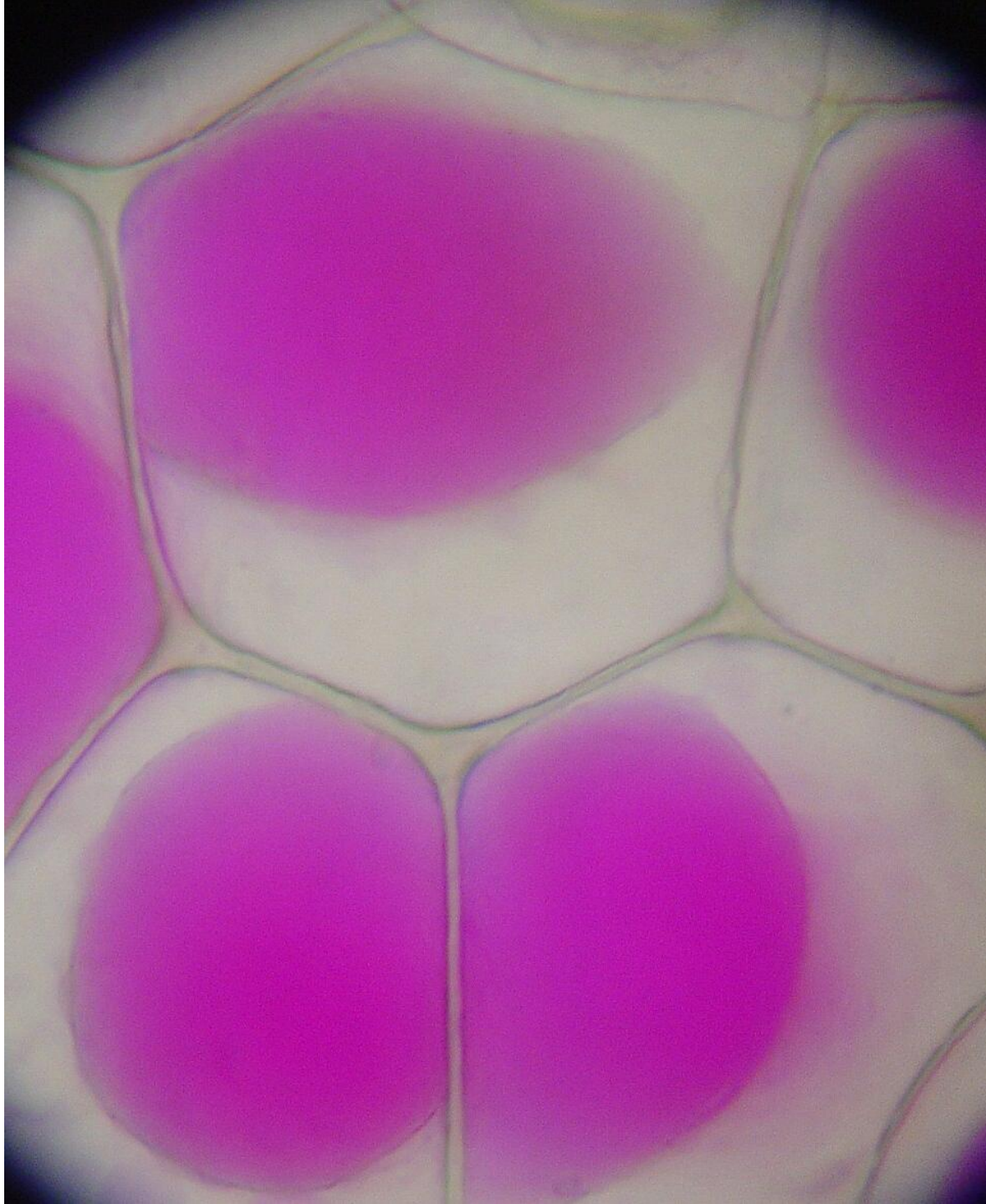


Image: plasmolysed cells of a leaf epidermis. Mnolf, CC BY-SA 3.0.

# Objectives for each class

| Class | Statements | By the end of the class, you can... |
|---|---|---|
| 1 | D2.3.8–9 | Define water potential and predict the direction of water movement |
| 2 | D2.3.10–11 | Use solute and pressure potential to explain water movement in plant tissue |

# What is water potential?



- **Water potential** measures the stored energy held by a given volume of water
- Symbol: the Greek letter psi; unit: kPa
- It is measured relative to pure water at standard pressure, which is set at **0 kPa**
- Dissolving a solute lowers water potential: solutions have **negative** values
- The more concentrated the solution, the more negative its water potential

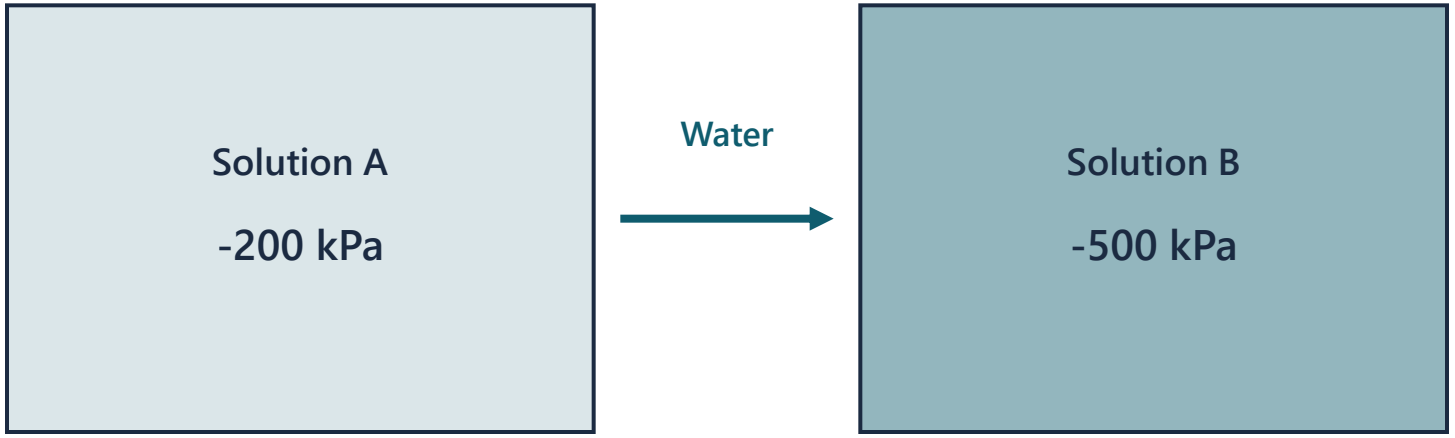
| Liquid | Water potential / kPa |
|---|---|
| **Pure water** | 0 |
| **Dilute solution** | about -100 |
| **Cell sap of a leaf** | about -1000 |
| **Sea water** | about -2500 |

### Why solutes lower it

Water molecules form hydrogen bonds with solute particles, so they are less free to move: their potential energy falls.

# Water moves down a water potential gradient

D2.3.9




Separated by a partially permeable membrane. Darker: more solute.

- Water moves from **higher** (less negative) to **lower** (more negative) water potential
- So water moves from A to B
- Movement stops when the water potentials are equal

## Try it

Three cells touch in a row: X at -300 kPa, Y at -500 kPa and Z at -400 kPa. Draw arrows to show the net movement of water between each pair.

# Solute potential and pressure potential

D2.3.10

$$\text{water potential} = \text{solute potential} + \text{pressure potential}$$

| | Solute potential | Pressure potential |
|---|---|---|
| **Caused by** | Dissolved solutes | Pressure of the cell contents on the cell wall |
| **Usual sign** | Zero or negative | Positive in turgid cells |
| **Gets lower when** | More solute dissolves | The cell loses water |
| **Note** | Pure water has 0 kPa | Negative in xylem, where sap is under tension |

**Key idea**

A cell wall lets pressure build up, so pressure potential matters most in plants, fungi and bacteria.

# Using the equation

D2.3.10

| Cell | Solute potential / kPa | Pressure potential / kPa | Water potential / kPa |
|---|---|---|---|
| **Turgid cell in pure water** | −700 | +700 | 0 |
| **Cell in a leaf on a sunny day** | -1000 | +400 | -600 |
| **Flaccid cell** | -900 | 0 | -900 |
| **Plasmolysed cell** | -1200 | 0 | -1200 |

### Try it

A cell has a solute potential of -800 kPa and a pressure potential of +300 kPa. Calculate its water potential.

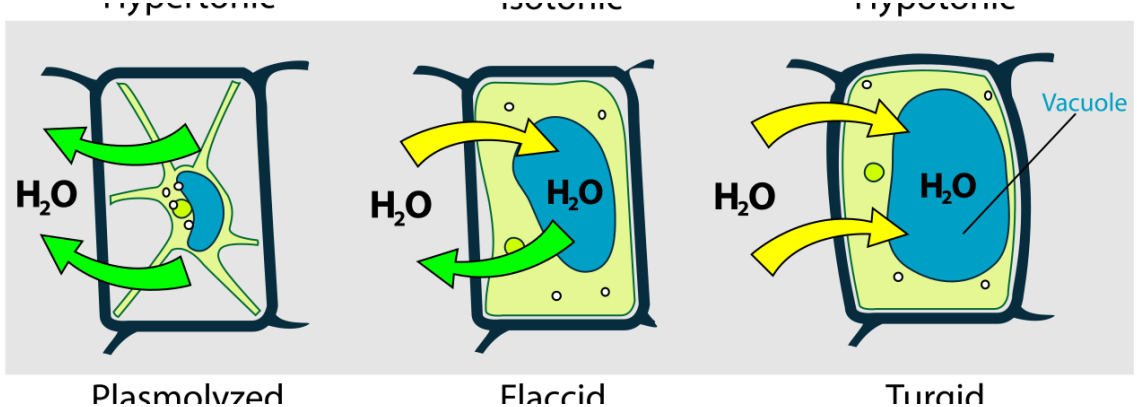
### Then

It is placed in a solution at -400 kPa. Does it gain or lose water?

# Plant tissue in a hypotonic solution

D2.3.11

Solution has a higher water potential than the cells → Water enters by osmosis → Cell sap becomes more dilute: solute potential less negative → **Protoplast presses on the wall: pressure potential rises** → Water potential rises until it equals the solution




Plant cells in hypertonic, isotonic and hypotonic solutions. LadyofHats, public domain.

## Result

The cell becomes turgid. The wall stops it bursting. Turgid cells hold up leaves and young stems.

# Plant tissue in a hypertonic solution

D2.3.11

Solution has a lower water potential than the cells → Water leaves by osmosis → Cell sap becomes more concentrated: solute potential more negative → Pressure potential falls to zero: cell is flaccid → Protoplast shrinks from the wall: plasmolysis



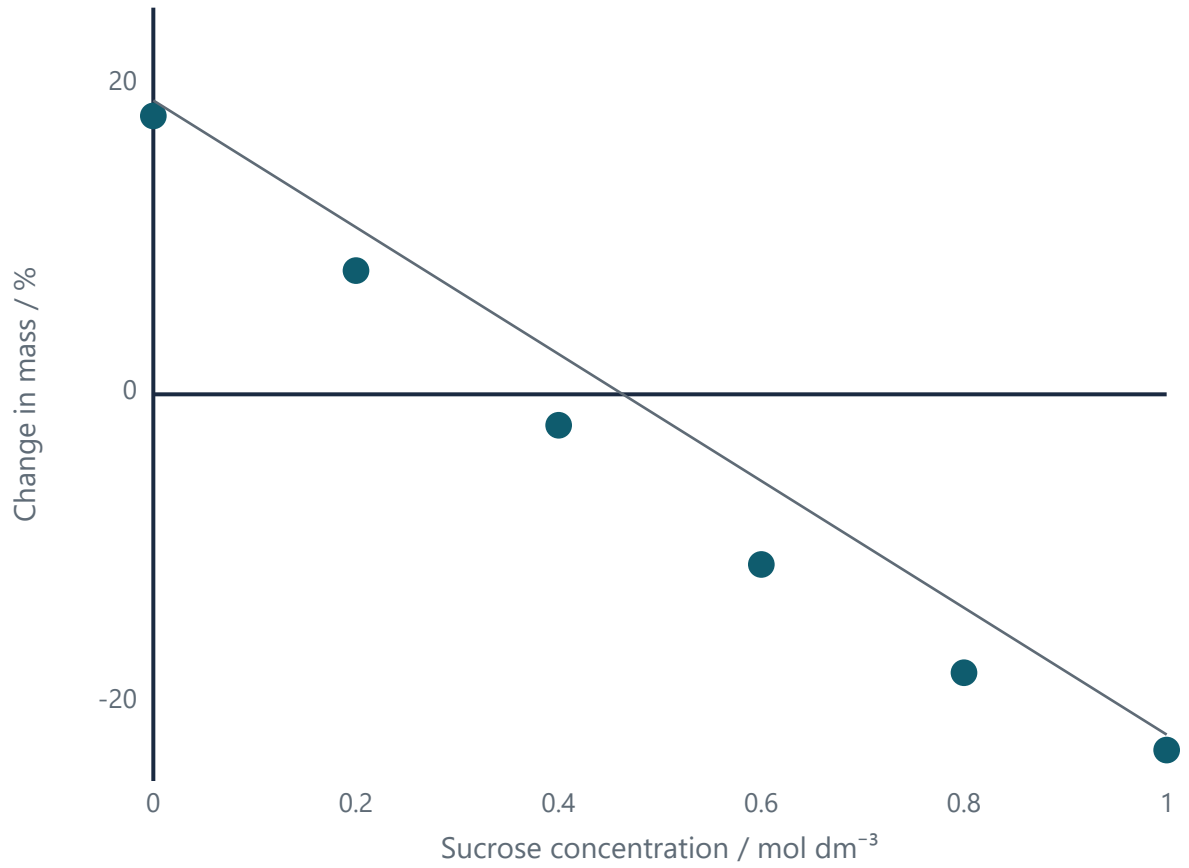
Plasmolysed epidermis cells. Mnolf, CC BY-SA 3.0.

## Why plants wilt

In drought or salty soil, cells lose water, pressure potential falls and leaves droop. Few plants survive in sea water: its water potential is far lower than that of most plant cells.

# Estimating the water potential of a tissue

D2.3.11




Invented results for potato cylinders after 24 h.

- Weigh tissue samples, leave them in a range of sucrose solutions, then reweigh
- Plot percentage change in mass against concentration
- Where the line crosses zero, there is no net water movement
- So the tissue has the same water potential as that solution

**Try it**

Read the concentration where the line crosses zero, then look up its water potential in a table.

# Going further: from soil to air



Water moves through a plant because water potential falls at every step from the soil to the air. These are typical values on a sunny day.

| Place | Water potential / MPa |
|---|---|
| Moist soil | -0.3 |
| Root cells | -0.6 |
| Xylem in the stem | -0.8 |
| Leaf cells | -1.5 |
| Air (50% humidity) | -95 |

**Think**

Why is the biggest drop between the leaf and the air? What would happen if the soil dried to -2.0 MPa?

# Check-in: water potential

D2.3.8–10

1. State the water potential of pure water at standard pressure.
2. State the direction water moves between two regions of different water potential.
3. Explain how adding solute affects water potential.
4. Explain how pressure potential changes as a plant cell takes in water.
5. A cell has a solute potential of -0.8 MPa and a pressure potential of 0.3 MPa. Calculate its water potential.
6. Explain how to find the water potential of a tissue from a set of sucrose solutions.

**How to use this**

Answer without your notes, then compare with a partner.

# Working with data: estimating tissue water potential

D2.3.10

Percentage change in mass of beetroot discs in sucrose solutions. These data are invented for practice.

| Sucrose / mol $dm^{-3}$ | Solute potential / MPa | Mass change / % |
|---|---|---|
| **0.1** | -0.26 | +8 |
| **0.2** | -0.54 | +3 |
| **0.3** | -0.82 | -2 |
| **0.4** | -1.12 | -7 |

1. Estimate the sucrose concentration at which mass does not change.
2. Estimate the water potential of the tissue.
3. Explain why repeats improve the estimate.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Water moves to a higher water potential.** | It moves to a lower (more negative) water potential. |
| **Solute potential can be positive.** | Solute potential is zero or negative. |
| **Pressure potential is negative in turgid cells.** | It is positive in turgid cells. |

# Check that you can...

1. **Define** water potential and **state** its units
2. **Explain** water movement using water potential
3. **Calculate** water potential from solute and pressure potential
4. **Explain** changes in plant tissue in hypotonic and hypertonic solutions
5. **Estimate** the water potential of a tissue from data

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. State the water potential of pure water.

2. A cell has a solute potential of -900 kPa and a pressure potential of +500 kPa. Calculate its water potential.

3. Explain what happens to the pressure potential of a cell placed in a hypertonic solution.

# Image credits

Rhoeo Discolor – Plasmolysis.jpg – Mnolf – CC BY-SA 3.0

Turgor pressure on plant cells diagram.svg – LadyofHats – Public domain

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